

Answer **all** questions.

1. (a) The table shows four compounds and their formulae.

| Compound         | Formula                       |
|------------------|-------------------------------|
| nitrogen dioxide | NO <sub>2</sub>               |
| hydrogen bromide | HBr                           |
| ethane           | C <sub>2</sub> H <sub>6</sub> |
| ammonia          | NH <sub>3</sub>               |

Use the information in the table to answer the following questions.

- (i) Give the
- name**
- of the compound that contains the elements hydrogen and bromine. [1]

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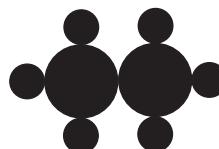
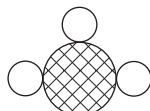
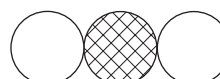
- (ii) Give the
- names**
- of the elements present in ethane. [1]

..... and .....

- (iii) Give the
- total**
- number of atoms found in a molecule of nitrogen dioxide. [1]

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- (iv) Tick (✓) the box next to the diagram that represents a molecule of ammonia, NH
- <sub>3</sub>
- . [1]


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- (b) Some ants spray a toxic mixture containing formic acid to defend themselves from predators.

Each molecule of formic acid contains two atoms of hydrogen, one atom of carbon and two atoms of oxygen.

Use this information to give the formula of formic acid.

[1]

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- (c) The box contains the names of three compounds that are made from positive and negative ions.

**lithium chloride**

**magnesium bromide**

**calcium oxide**

- (i) Give the formulae of the **ions** present in lithium chloride.

[1]

..... and .....

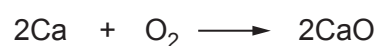
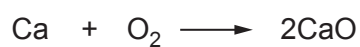
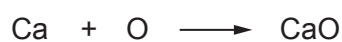
- (ii) Give the formula of magnesium bromide.

[1]

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- (iii) Tick (✓) the box next to the correct equation for the reaction between calcium and oxygen to produce calcium oxide.

[1]


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